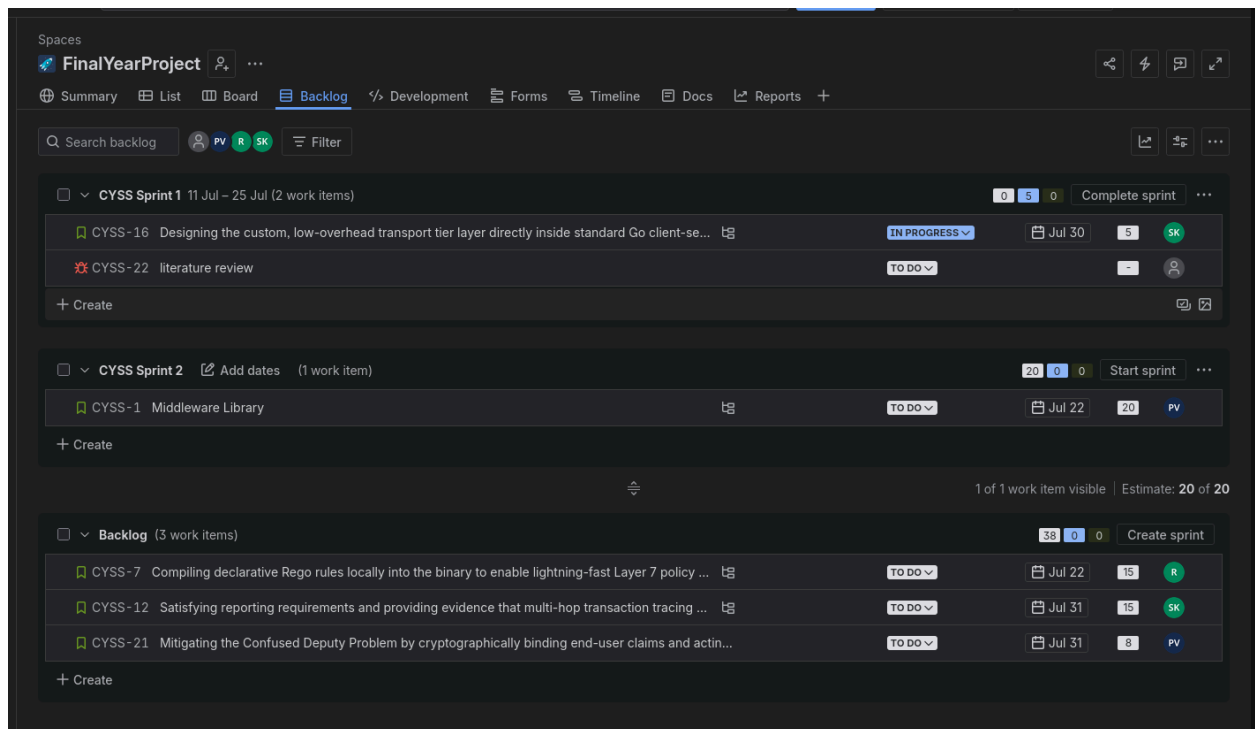
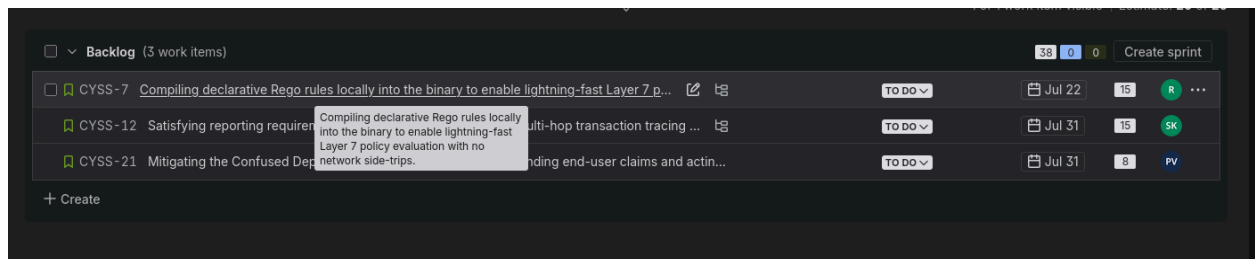


Praneesh R V
CB.SC.U4CYS23036

Scrum



Backlog



These tasks are waiting to be assigned and placed in sprints

With each story having subtasks

Spaces / FinalYearProject / Add epic / CYSS-7

Compiling declarative Rego rules locally into the binary to enable lightning-fast Layer 7 policy evaluation with no network side-trips.

✓ **Description**

As a **Security Administrator**, I want incoming microservice network requests to evaluate access control rules locally using an embedded OPA engine, so that fine-grained Layer 7 authorization checking happens entirely in-memory with zero network-hop latency

✓ **Subtasks** ... +

0% Done

Work	Prior...	Assi...	Status
CYSS-8 Import and initialize the Open Policy Agent (OPA) Go SDK package directly into the library footprint	= Me	R	TO DO ▾
CYSS-9 Design a middleware interface wrapper that intercepts incoming requests to harvest L7 context payload strings...	= Me	Un	TO DO ▾
CYSS-10 Implement a local system bundle loader that reads declarative .rego rule policy files compiled into the binar...	= Me	Un	TO DO ▾
CYSS-11 Write benchmark validation tests comparing local embedded OPA decision evaluations against external...	= Me	R	TO DO ▾

Linked work items

Add linked work item

They are assigned to users and story points are mentioned

Satisfying reporting requirements and providing evidence that multi-hop transaction tracing works.



To Do ▾

 Agents



 Improve Story

✓ Description

As a **Compliance Auditor**, I want the framework to log structured, cryptographically verified multi-hop user contexts for every network request chain, so that I can easily reconstruct an unalterable audit trail tracing which user authorized specific downstream behaviors.

✓ Subtasks

...  +

0% Done

Work	Pri...	As...	Status
 CYSS-13 Integrate high-performance structured logging formatters (su...	= M	 U	TO DO ▾
 CYSS-14 Standardize log schema structures to output JSON records explicitly...	= M	 U	TO DO ▾
 CYSS-15 Create a multi-service loop telemetry trial run verifying that...	= M	 U	TO DO ▾

Linked work items

 Add epic /  CYSS-1



 2





Middleware Library





To Do ▾

 Agents



 Improve Story

▼ Description

As a **Microservice Developer**, I want the middleware library to automatically connect to a local SPIRE agent at startup and retrieve an X.509 SVID certificate, so that I don't have to manually provision, rotate, or manage cryptographic keys

▼ Subtasks

...  +

0% Done

Work	Pri...	As...	Status
 CYSS-3 Implement a Unix Domain Socket (UDS) connection client in Go to...	= M	 F	IN PROGRESS ▾
  CYSS-4 Write a parsing module to extract the SPIFFE ID URI (e.g., spiffe://... 	= M	 F	IN PROGRESS ▾
 CYSS-5 Create a background go-routine channel listener that monitors the...	= M	 F	IN PROGRESS ▾
 CYSS-6 Write integration tests utilizing a mocked SPIRE Agent to verify...	= M	 F	IN PROGRESS ▾

Linked work items

Add linked work item

Confluence content

... +

 Product requirements

TRY TEMPLATE

✓ Jira work item

↗

×

🔗 Add epic

/

🔖 CYSS-16

🔒

👁

🔗

⋮

Designing the custom, low-overhead transport tier layer directly inside standard Go client-server network components

+

⋮

In Progress ▾

🏠 Agents

⚡

⬆️ Improve Story

▽ Description

As a **Security Administrator**, I want all internal service-to-service communication channels to enforce strict in-process mutual TLS handshakes using live SVIDs, so that unauthorized network requests are rejected instantly at the transport layer without heavy sidecar proxy hops.

▽ Subtasks

⋮

📅

+

50% Done

Work	Pri...	As...	Status
🔗 CYSS-17 Construct a custom tls.Config generator that maps the live...	==	M	SK Ⓢ DONE ▾
🔗 CYSS-18 Build a wrapper factory for standard http.Server and http.Clie...	==	M	SK Ⓢ DONE ▾
🔗 CYSS-19 Write a handshake interceptor rule that forces mutual verification...	==	M	SK Ⓢ IN PROGRESS ▾
🔗 CYSS-20 Write a network execution test verifying that connections...	==	M	SK Ⓢ TO DO ▾

Linked work items

Add linked work item

Confluence content

⋮

+

Sprint 1 - time period, goal, tasks

Edit sprint: CYSS Sprint 1



Required fields are marked with an asterisk *

Sprint name *

CYSS Sprint 1

Start date *

7/11/2026



3:37 PM



End date *

7/25/2026



3:37 PM



Automatically complete sprint

Sprint goal

Designing the custom, low-overhead transport tier layer directly inside standard Go client-server network components

Cancel

Update

Spaces / FinalYearProject / Add epic / CYSS-16

Designing the custom, low-overhead transport tier layer directly inside standard Go client-server network components

Description

As a **Security Administrator**, I want all internal service-to-service communication channels to enforce strict in-process mutual TLS handshakes using live SVIDs, so that unauthorized network requests are rejected instantly at the transport layer without heavy sidecar proxy hops.

Subtasks

50% Done

Work	Prior...	Assi...	Status
CYSS-17 Construct a custom <code>tls.Config</code> generator that maps the live memory-buffered SVID assets to standard Go structures.	= Me	SK S	DONE
CYSS-18 Build a wrapper factory for standard <code>http.Server</code> and <code>http.Client</code> (plus <code>grpc.Server</code> and <code>grpc.ClientConn</code>) to inje...	= Me	SK S	DONE
CYSS-19 Write a handshake interceptor rule that forces mutual verification (<code>tls.RequireAndVerifyClientCert</code>).	= Me	SK S	IN PROGRESS
CYSS-20 Write a network execution test verifying that connections originating from workloads without a valid SPIRE identit...	= Me	SK S	TO DO

Linked work items

Add linked work item

Agents

Start work

Uses AI. Verify results.

Add a comment...

In Progress

Agents

Improve Story

Details

Assignee Sanjai K C K
[Assign to me](#)

Parent None

Sprint [CYSS Sprint 1](#)

Labels None

Due date Jul 30, 2026

Team None

Start date Jan 16, 2026

Story point estimate **5**

Reporter Sanjai K C K

Development

Automation

Recent rule runs

Refresh to see recent runs.
[Create new automation rule](#)

Created yesterday
Updated 1 hour ago

Refresh

Confirm

Spaces

FinalYearProject

Summary

List

Board

Backlog

Development

Forms

Timeline

Docs

Reports

Search timeline

PV R SK

Filter

Work

Sprints

[CYSS-16](#) Designing the custom, low-overhead tr...

SK Sanjai K ...

[CYSS-1](#) Middleware Library

PV Praneesh...

[CYSS-7](#) Compiling declarative Rego rules locally ...

R RadKriss

[CYSS-12](#) Satisfying reporting requirements and p...

SK Sanjai K ...

[CYSS-21](#) Mitigating the Confused Deputy Proble...

PV Praneesh...

[CYSS-22](#) literature review

Unassign...

+ Create

6 of 6

July

August

CYSS Sprint 1

Activity Board

Spaces

FinalYearProject

SummaryListBoard

logDevelopmentFormsTimelineDocsReports

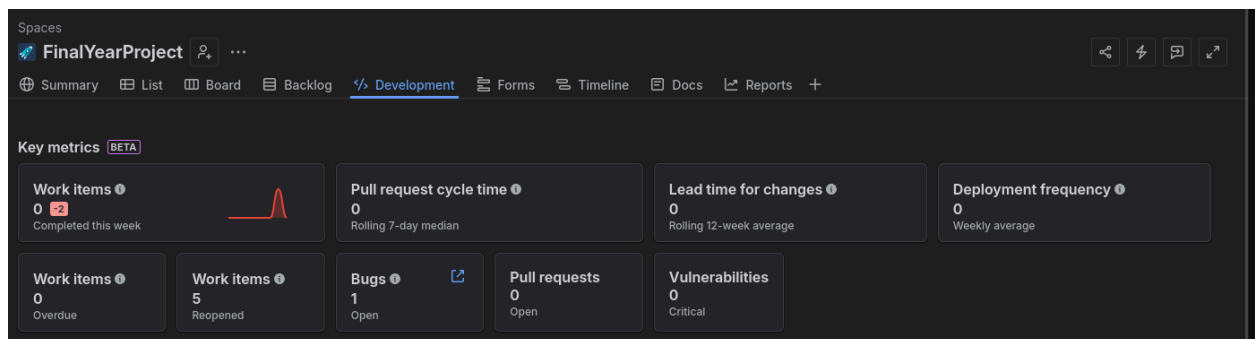
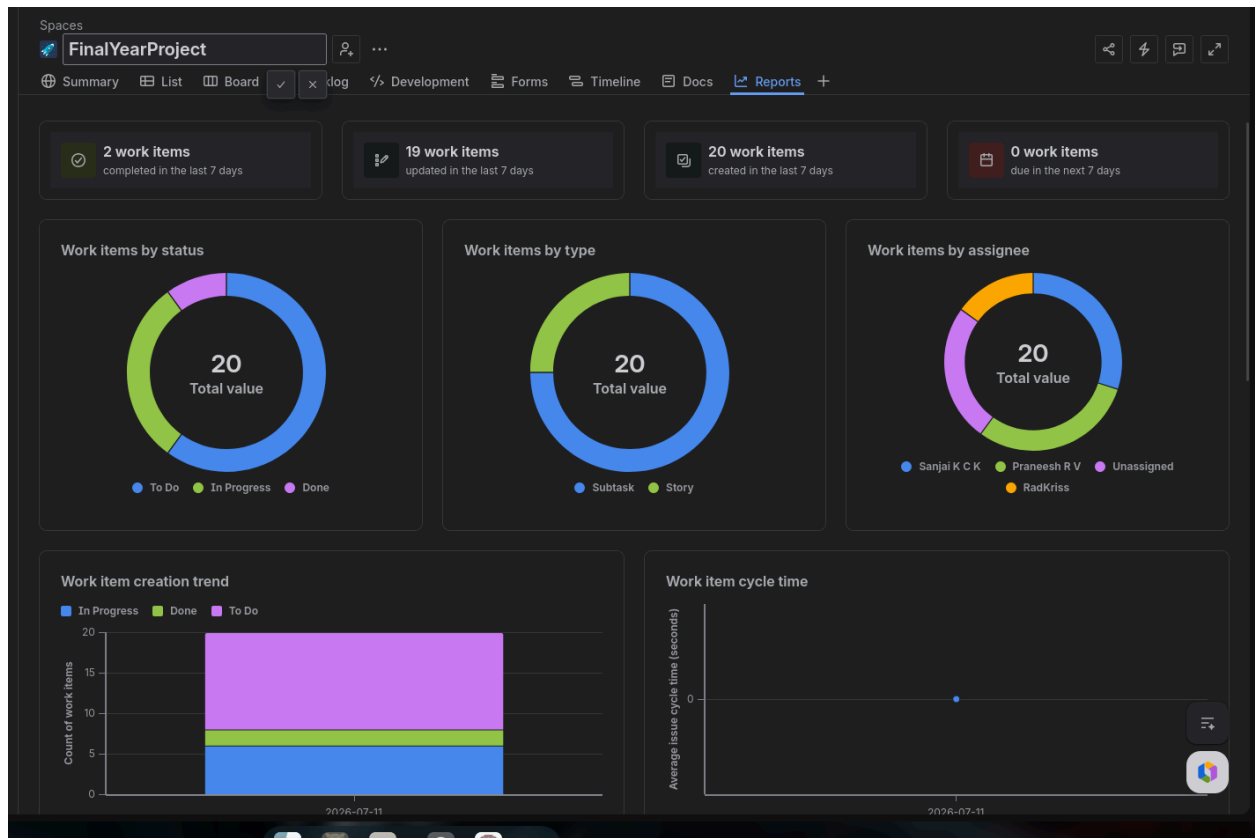
Ask AISearch workFilterGroup

Work	Assignee	Reporter	Priority	Status	Resolution	Creator
CYSS-16 Designing the custom, low-overhead transp...	SK Sanjai K C K	SK Sanjai K C K	Medium	IN PROGRESS	Unresolved	Jul 11, 2026
CYSS-17 Construct a custom tls.Config generator th...	SK Sanjai K C K	SK Sanjai K C K	Medium	DONE	Done	Jul 11, 2026
CYSS-18 Build a wrapper factory for standard http.S...	SK Sanjai K C K	SK Sanjai K C K	Medium	DONE	Done	Jul 11, 2026
CYSS-19 Write a handshake interceptor rule that for...	SK Sanjai K C K	SK Sanjai K C K	Medium	IN PROGRESS	Unresolved	Jul 11, 2026
CYSS-20 Write a network execution test verifying th...	SK Sanjai K C K	SK Sanjai K C K	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-1 Middleware Library	PV Praneesh R V	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-3 Implement a Unix Domain Socket (UDS) co...	PV Praneesh R V	PV Praneesh R V	Medium	IN PROGRESS	Unresolved	Jul 11, 2026
CYSS-4 Write a parsing module to extract the SPIF...	PV Praneesh R V	PV Praneesh R V	Medium	IN PROGRESS	Unresolved	Jul 11, 2026
CYSS-5 Create a background go-routine channel lis...	PV Praneesh R V	PV Praneesh R V	Medium	IN PROGRESS	Unresolved	Jul 11, 2026
CYSS-6 Write integration tests utilizing a mocked S...	PV Praneesh R V	PV Praneesh R V	Medium	IN PROGRESS	Unresolved	Jul 11, 2026
CYSS-7 Compiling declarative Rego rules loc...	R RadKriss	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-8 Import and initialize the Open Policy Agent...	R RadKriss	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-9 Design a middleware interface wrapper tha...	Unassigned	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-10 Implement a local system bundle loader th...	Unassigned	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-11 Write benchmark validation tests comparin...	R RadKriss	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026
CYSS-12 Satisfying reporting requirements and provi...	SK Sanjai K C K	PV Praneesh R V	Medium	TO DO	Unresolved	Jul 11, 2026

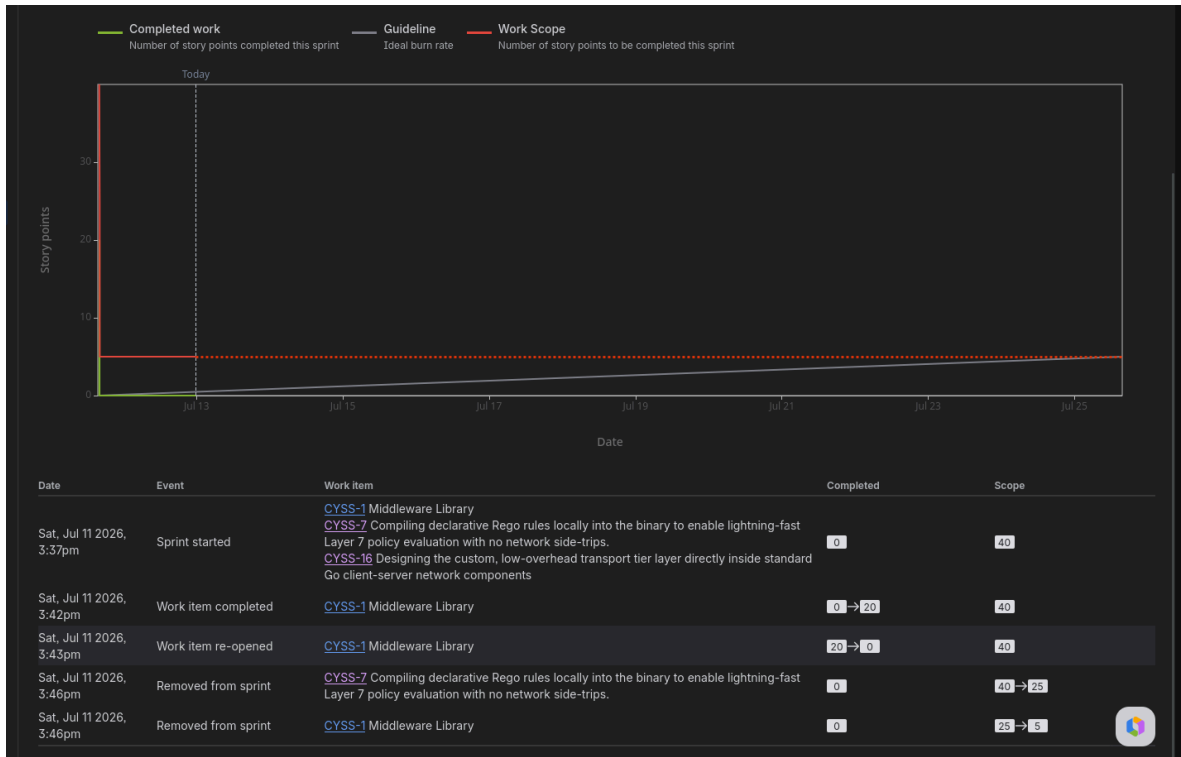
+ Create

5 of 5

Metrics



Burnup Chart



Burndown Chart

